OPEN-SOURCE BUSINESS MODELS



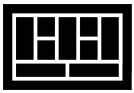
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SUPER GUIDE:

OPEN-SOURCE BUSINESS MODELS

BY DANIEL PEREIRA



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INTRODUCTION

The open-source business model, as used by many tech-driven organizations, is currently not well understood. Although open-source is built free with its source code available for everyone to use and tinker with, it still generates revenue amounting to millions, even billions of dollars. Hence, its successes still remain a mystery to many. This model has been used to build multi-billion-dollar companies, like Confluent ($4.5B), HashiCorp ($5.3B), Databricks ($6.2B), Elastic ($9.3B), and MongoDB ($13.6B). How did these companies utilize the open-source business model to grow from free projects to multi-billion-dollar companies? This super guide on open-source business models will help you to understand the core principles of open-source and how to apply them in real-life situations. It will guide you through the process of implementing the model, take away any guesswork, and help you to focus on the creation, delivery, and capture of values.

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WHAT IS OPEN-SOURCE?

Open-source, sometimes referred to as Open-Source Software (OSS), is a software program publicly distributed with its source code, making it accessible for anyone to modify and distribute as they deem fit. A source code is the set of instructions and statements written and manipulated by computer programmers, using a computer programming language, to control how a program or application behaves.

When a source code is accessible by other programmers, they can modify its programs by adding to it for enhancement, changing it to fix an error, or simply manipulating it to fit their own needs. Open-Source Software (OSS) is usually developed in a collaborative way, relying on communities of developers and users. Typically, OSS includes a license that determines the way people can use, study, modify, and distribute the software. For example, a license may stipulate that anyone who alters and shares an OSS program with others must also share that program's source code without charging a licensing fee for it.

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WHY DOES OPEN-SOURCE MATTER TODAY?

You might naturally think it is about money, standards, or interoperability, but the most important element behind the popularity of open-source today is its capability to offer control to everyone.

By its design, open-source promotes a collaborative exchange of ideas within a creative community in order to enhance the advancement of innovations and technology. Though most of its application is within the software industry, it has found applications in other industries, such as electronics, robotics, biotech, etc.

Open-source provides an avenue for companies to collaboratively develop technologies like software products. They develop the software together and then publish the source code so that it can be accessed by anyone to test it, replicate it or make suggestions on how to make it better. Though it feels more natural for them to keep the source code secret and make more money from it, making it publicly available provides a standardized and leveled playing field for everyone upon which more specialized versions can be built.

Innovation is one of the best things about open-source. The open nature of open-source ensures that its strong community is able to enhance the product by building custom-fit solutions based on its already existing code. The new solution is open-sourced and made available for public

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use. The speed of innovation on open-source software enables faster response to market and individual needs.

Open-source affords developers immense control and flexibility, allowing them to “stand on the shoulders of giants” to build their own software.

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OPEN-SOURCE ORIGIN STORY

In the 1950s and 1960s, researchers working on the development of the early internet, telecommunication technologies, and protocols relied on open and collaborative efforts. These technologies, which later became the foundation of the internet, were a result of peer review and an open feedback process. By the time of the birth of the internet in the 1990s, the values of peer review, collaboration, and openness to development projects were written into its foundation.

By the end of the 1990s, “open-source” started as a software licensing model, where the code is accessible by everyone for modification and distribution under the open-source license. This gave rise to the business model upon which many companies have grown into multi-billion-dollar enterprises. The release of the Netscape browser’s source code and the recognition of Linux in mainstream media increased participation within the open-source community.

The Open-Source Initiative (OSI) was formed in 1998 as an educational, advocacy, and stewardship organization to foster the growing industry of collaborative development.

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COMMERCIAL USE OF OPEN-SOURCE

The commercial use of open-source software implies that the software has been licensed to a commercial company under an open-source license. The company may make use of the open-source as it deems fit, but must also fulfill its obligations as contained in the license.

A commercial company may get an open-source software supplied by either an open-source community or another commercial company. The open-source community — a community of people that builds, maintains, and supports the open-source software most often for free of charge while Commercial Open-Source Software (COSS) — refers to a category of companies that commercialize open-source technology, which requires a paid license, bound by the terms of a company's license.

The differences between them determine the important decision of choosing one of them for your project. We shall cover the differences in a later section.

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IS OPEN-SOURCE A DEVELOPMENT MODEL OR A BUSINESS MODEL?

Software development models provide the framework used to plan and execute software milestones and delivery cycles throughout the life of a software, while a business model is how a company creates, delivers, and captures values.

Open-source as a developmental model emphasizes its collaborative and decentralized system of building its code and delivery of its license. On the other hand, open-source as a business model emphasizes its monetization through its paid features, services, support, and marketing.

Both models hold true to an extent on open-source software, but neither has satisfied everyone. When considered within the historical context of their practical construction, some think of open-source as a supply chain where technology can be purchased.

A typical supply chain comprises organizations, people, activities, and resources needed for building and delivering any product or service. Similarly, all of the components of a supply chain are obtainable in open-source software, and they are keys to its success.

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OPEN-CORE VS. HYBRID BUSINESS MODELS

The source code and the business model are often a cause of confusion for people.

The open-core is a business strategy in which a company offers a ‘core’ or feature-limited version of a product as free and open-source, while offering an add-on commercial version as proprietary software. Hence, the resulting open core is a combination of open-source and closed source attributes, which could be referred to as freemium software.

The free version of the product is usually built on an open-source, while the company develops an enterprise version in areas where the project is deficient. These added features are aimed at paying clients.

The Hybrid business model is a slightly different approach, which aims to combine the best worlds of open-source and proprietary software by building proprietary products and services on collaboratively developed open-source platforms. This model offers several advantages over the open core model. It builds applications or services on open-source platforms. The company delivers these applications and services as it deems best, either as proprietary, open-source, or even open core, but the platform from which they were built remains fully open-source and open to collaborators.

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PRODUCT OR PROJECT?

A product is an item or service people buy with currency to serve their needs or wants, while projects are events people participate in or contribute to. Open-source can be consumed as a product or project.

- As a product, a vendor selects an open-source, compiles it, tests it, and then sells support for the package, which usually includes docs, patching, and updates;
- As a project, anyone selects an open-source, compiles it, tests it, and then manages it.

Anyone can download and use open-source however they choose, but as soon as a product built upon an open-source is sold, then the responsibility is owed to the customer. The seller of the product has a commitment to satisfy the customer.

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OPEN-SOURCE VS. FREEMIUM: OPEN-SOURCE BUSINESS MODEL

- Open-source is usually developed, licensed, and maintained by a community of independent developers, while freemium is developed and distributed by a vendor;
- In an open-source model, the vendor distributes the product or service as per its open licensing model, while in freemium, the vendor has full control over its distribution.

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COMMERCIAL OPEN-SOURCE VS. COMMUNITY OPEN-SOURCE

Commercial open-source is owned and managed by a legal entity for the purpose of revenue generation, while a community open-source is owned by a community of independent developers.

Commercial open-source requires a paid license guided by the terms of the company's license, while a community open-source is free and accessible by the public and bound by the terms of the community license.

Commercial open-source provides add-on features that require payment from customers to access it, while the open-source community is free but involves other hidden costs like time invested to fix any issue within the software.

Commercial open-source offers guaranteed support on improvements and updates, while the open-source community may suffer from poor participation by community members to respond to needed upgrades.

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OPEN-SOURCE SOFTWARE VS. FREE SOURCE SOFTWARE

Although both terms refer to a similar set of licenses and software, they possess different underlying ideologies.

The term ‘free software’ has nothing to do with price. It is about freedom! It implies that users have the freedom to run, copy, distribute, study, change, and improve the software.

Free software allows users the ultimate freedom to do anything they want with it, like even improving the version and profiting from it.

Open-source emphasizes the practical benefits of “free software” by supporting the engagement of users and developers to create and improve source codes through collaboration on software development projects.

Hence, it can be said that open-source is a business-oriented developmental philosophy while free software is a social cum moral philosophy.

Examples of free software include the Apache web server and the Linux kernel.

Examples of open software include Mozilla Firefox and LibreOffice.

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CLASSIFICATION OF OPEN-SOURCE BUSINESS MODELS

Though open-source software is offered free of charge, there is usually some form of compensation accruing to the open-source component. There are two business models worth considering:

- Community open-source business model
- Commercial open-source business model

COMMUNITY OPEN-SOURCE BUSINESS MODEL

The community open-source business model consists of the following business models:

- Inventor
- IP Lessor
- Contractor

The inventor business is carried out by the community. The community coordinates the work and gathers contributions from members to create the open-source software.

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The IP Lessor business deals with the terms and conditions of the open-source license, guiding how customers make use of the software. The community defines the license, which binds all customers to its terms and conditions. A customer may have the liberty to choose from multiple licenses of open-source software.

The contractor business comprises services like maintenance, support, and language translation. They are usually carried out by community members with zero charges to the customer. The customer typically has little or no control over these services since there are no service level agreements.

COMMERCIAL OPEN-SOURCE BUSINESS MODELS OVERVIEW

The commercial open-source business model allows companies to capture part of the value that is delivered by their open-source core product. These companies capture value through offering commercial versions of their products, add-on proprietary software, or alternative license options whilst still depending on their open-source core software.

Companies leverage open-source for their business models by:

- Using the open-source community as a supplier of software products: Most commercial software in the market today contains parts that are under an open-source license. Open-source affords them quality and cost advantages since they can get the software for free;

- Providing software to customers under a commercial

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license: Commercial vendors typically add value to open-source software before offering it to customers. They add value through creating a distribution chain or creating customer-tailored adaptations of the open-source software;

- Providing services around open-source software to customers: Such services include: 1) Maintenance; 2) Support; 3) Implementation; and 4) Adaptation of open-source software.

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MOZILLA OPEN-SOURCE MODEL

Mozilla provides the Firefox web browser as an open-source web browser originally developed by the Mozilla foundation. The Mozilla open-source software is held by Mozilla's foundation. Though the Mozilla browser is free, its revenue comes through deals with search engines to be featured as default on its browser, and via its subscription service.

In 2018, the Mozilla Corporation generated $435.702 million from royalties mostly coming from a deal with Google, which featured as a default search engine on the Mozilla browser.

Its subscription revenue comes from its pocket premium, which is a premium subscription enabling advanced features like full-text search on saved articles, removal of advertising, highlights, and more.

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SLACK FREEMIUM MODEL

SLACK (Searchable Log of All Communication and Knowledge) is a cloud-based team messaging application that enables team members to communicate via chat, phone, or video. Users can form groups called channels to collaborate on a project or exchange information. Slack can be accessed on both desktops and mobiles.

Slack uses a freemium model, where a free version is offered with limited features. Users can upgrade to a paid service if they need to unlock advanced features and functionality.

Slack generates revenue primarily from paid subscriptions. Its subscriptions are paid on a monthly and annual basis.

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THE 5 OPEN-SOURCE BUSINESS MODELS

Open-source companies have successfully commercialized their products through five common models:

1. Support services
2. Hosting
3. Restrictive licensing
4. Open core
5. Hybrid licensing

SUPPORT

Most open-source companies actually started from the support model. The model entails the sale of deployment and integration services, training, production-oriented insurance policies, bug fixes, certified binaries, etc., to customers willing to deploy the project in production.

This model is well known for its inefficiency due to its limit over a long period of time. These limits include:

- The labor-intensive nature of the support due to its reliance on manual work;
- Repetition in support work is often not repeatable,

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hence, scaling becomes hard;

- It is only profitable if the project requires complex deployments with a sprawling ecosystem, hence, making the project easier destroys the revenues from support.

Over time, customers who rely on such projects often invest in their own engineering capacity to understand and support the project themselves instead of relying on external support.

HOSTING

The hosting model entails offering a fully managed version of your project, such that users can simply access a remote server with the software with just a few clicks. With this model, users do not need to worry about backups, upgrades, or downtime.

The hosting model is very popular with public cloud providers and cloud-based companies, as it allows them to monetize their open-source projects. The hosting model is often layered with a few other models.

RESTRICTIVE LICENSING

The restrictive licensing model comes with arduous terms such that using the software in production is incentivized to enter a deal with the vendor. Hence, the restrictive licensing model invents legal grounds for users of open software to pay.

A major disadvantage of this model is that the arduous terms of its licenses hurt adoption and are a major turn-off for potential users.

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The GPL and AGPL licenses, as well as the newly created Commons Clause (adopted by certain Redis modules), are examples of this model. The GPL-based license restrictions only apply if a user makes modifications and does not want to open-source them.

OPEN-CORE

The Open core model is one in which a company offers a ‘core’ or feature-limited version of a product as free and open-source, while offering an add-on commercial version as proprietary software.

The majority of the code is open-source, while a smaller percentage (known as the core and targeted at enterprise users) is proprietary.

The proprietary version is the one needed for deployment and scale. It lets an open-source company license the core permissively while retaining the ability to charge for the proprietary features. The company is able to keep valued features in its proprietary code base away from free-loading users like public cloud providers.

To use this model successfully, the company must be able to balance the open-source value versus the proprietary value. If it gives away too much value, it reduces its ability to make money; but if it offers too little value, then the open-source becomes ‘lame-ware’.

Another challenge from an engineering perspective is that separation and maintenance of two different code bases could be tough. For example, it is a challenging engineering task to manage independently versioned releases that might need to interoperate and/or port code back and forth to prevent code divergence over time.

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Despite the challenges, It is the most popular model for open-source companies to monetize their offerings.

HYBRID LICENSING

The hybrid licensing is an improvement on the open core model. This model mixes up both the open-source and proprietary software in the same repository, and then makes the code for the entire repo available. This implies that the entire repository becomes open code, but not all are licensed under an OSI-approved open-source license.

Users can choose to use a binary with just the open-source bits (available under an open-source license), or use a binary with both the open-source and proprietary bits (available under the proprietary license). Typically, the proprietary licensed binary can only be unlocked by purchasing a license key.

The advantages of the hybrid model include all of the advantages of the open core model, including some extras, such as:

- It has a simpler engineering process and development since everything is contained in the same code base;
- It provides a common-place for the whole team to work on;
- It allows in-situ upgrades from free to paid versions without the usual down-time and salesman interaction;
- It allows an external contribution to proprietary features using the same workflow they use often.

Its biggest challenge is the same as that of open core, i.e. balancing the open-source value versus the proprietary value.

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HOW OPEN-SOURCE SOFTWARE TOOK OVER THE WORLD

Most of the software deployed by the world's biggest companies is open to the public. Anyone can download the source code behind Google's Android system or even Facebook's user interface.

How did we even get here, where the giving away of valuable information for free has suddenly taken over the world?

Back in the 1940s, when only a privileged few had access to binary programmable computers, the collaborative development of software also began. The 1960s witnessed the established practice of the modification and redistribution of source code, or the human-readable code base.

FROM OPEN-SOURCE TO OPEN CORE TO SAAS

The original open-source projects were not businesses, but were borne out of revolutions against the unjust profits accruing to proprietary software companies like Microsoft, Oracle, and SAP. Hence, beginning with operating systems and databases, progressive developers collaborated often asynchronously to build alternative software.

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Although the software was not user-friendly as they were made by developers for developers, it worked for the purpose it was made for, and it was flexible. The merits of this kind of software trickled across the software world, and over a decade, Linux became the most popular OS for servers (next to Windows), while MySQL rivaled Oracle's dominance.

Then came the attempt by entrepreneurs to adopt it into enterprise-grade support subscriptions. Red Hat emerged as the winner in the Linux race and MySQL (the company) for databases. The successful adoption of Linux and MySQL laid the foundation for the second generation of open-source companies like Cloudera and Hortonworks.

These second generation of open-source projects and businesses differed from the first generation in two major ways. Firstly, software was built by the employees of existing companies, and not by a community of independent developers. Secondly, the model they employed allowed only some part of the project to be licensed for free, while they charged a fee for the part under enterprise license.

The challenges faced by the second generation model included cut-throat competition for profits by offering increasing parts of the software for free, and frequent Balkanization of the evolution of the software in a bid to differentiate themselves.

The most recent evolution came with a better understanding of the business challenges faced by the two previous generations of open-source companies. The first lesson was that open-source software was now largely developed in-house by employees of companies that commercialized the software. The second one was that these companies from the onset now offer their software as a cloud service. Essentially, these are open core/cloud service hybrid businesses with various means to make money from their

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products.

Offering their products as Software-as-a-Service (SaaS) affords them the ability to interweave open-source software with commercial software, such that customers do not need to worry over which license they should be taking. Examples of this 3rd generation include companies like Mongo and Elastic.

This latest evolution implies that open-source software companies have the opportunity to thrive as the dominant business model for software infrastructure.

THE ROLE OF THE COMMUNITY

The open-source community plays a pivotal role in the development of open-source projects. The community tends to discover very innovative and relevant software and try them out. If it proves to be a good one, they often tend to give it a good review for other members to also try it out. These actions could make such a product trend within the community, hence, increasing its broad usage and acceptability.

Members of the community often take up the role of “management” by identifying problems with the product and seeking enhancement and improvements. They also help identify bugs and other shortcomings in the software, test 0.x versions, and give feedback to the companies. These give the open-source project an opportunity to transform itself into one with features and capabilities wanted and desired by developers.

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RISE OF THE DEVELOPER

The developers are the real customers of open-source software. The developers often download and integrate the software into the prototype versions of their own projects. With the integration of such open-source software into the projects of developers, the projects go through the complete cycle of development up to production. At this point, the open-source software can hardly be displaced.

The adoption of open-source software by developers often helps software diffuse through the experts and makes the selection process more basal than it has ever been.

VIRTUES OF THE OPEN-SOURCE BUSINESS MODEL

In comparison to closed source software companies, open-source software companies offer a lower cost per unit and are able to sell more units because it is more affordable for most people and enjoys quick and massive adoption.

Due to the massive adoption of open-source software, most customers already use instances of software in their environment. Hence, companies can easily market to customers to upgrade to the enterprise edition and get additional features. This shortens sales cycles and helps cut down costs of sales and marketing.

The large-scale adoption of open-source software allows for open-source software companies to be more efficient on a cash consumption basis, in comparison to traditional software companies.

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OPEN-SOURCE TO FREEMIUM

The community and customer base of open-source software have become more knowledgeable in recent times, and now appreciate the fact that open-source companies deserve some form of payment for their products so that they can continue to build and innovate.

Freemium is often recognized as the free use of software until it is deployed in production or scaled. Some companies, such as Elastic and Cockroach Labs, are open sourcing all their software but implementing a commercial license to parts of the software code. The reason is that they believe that real enterprise customers will pay whether the software is open or closed.

This movement has been largely facilitated by the migration of companies to licenses that are more business-friendly. Examples of such licenses include Mongo, Elastic, and Cockroach.

THE FUTURE

The potential of this model has been proven by companies now worth billions of dollars. Microsoft, which was one of the strongest proponents of proprietary technology and purveyor of open-source, said it has learned a lot from its increased engagement with the open-source world, adding that open-source is also now the "industry-accepted model for cross-company collaboration" and that it is now "all-in on open-source."

The 2022 Synopsys OSSRA (Open-Source Security and Risk Analysis) report found that 78% of codes in codebases are open-source and 97% of codebases contained open-source.

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These trends point to one single piece of information — more open-source. Open-source is set to be present in just about everything around technology.

“A few years ago, if you wanted to get several large tech companies together to align on a software initiative, establish open standards, or agree on a policy, it would often require several months of negotiation, meetings, debate, back and forth with lawyers… And did we mention the lawyers?,” wrote Sarah Novotny, Microsoft’s open-source lead for the Azure Office of the CTO.

Clearly, the growth trend of open-source software is undeniable.

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THE MOST POPULAR TYPES OF OPEN-SOURCE SOLUTIONS

Below are some of the most popular types of open-source solutions:

OPEN-SOURCE CRM

Customer Relationship Management (CRM) software improves customer care by facilitating the easy handling of customers' interactions and helping to meet their requirements easily. Happy customers mean better business for companies.

CRM software helps companies to stay organized and increases productivity. Using CRM software often requires some level of technical ability.

There is a variety of open-source CRM solutions available, which includes:

Hubspot CRM
Hubspot's main features include meeting management, contact management, email scheduling, and live chat.

It is best suited for small and medium businesses in retail, real estate, construction, and a host of other sectors.

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Flowlu
Flowlu helps business processes flow smoothly.

It is suitable for small and large businesses in a multitude of industries.

Its key features include project management, project workflows, invoicing, lead management, API integration, quoting, and many more.

Freshworks
Freshworks helps organizations to boost customer engagement and accelerate revenues with context-driven sales. It lets companies manage contacts, and develop customer success strategies.

It is suitable for businesses of all sizes.

Some of its key features include email marketing, phone, and email support, lead management, and integration management.

CiviCRM
CiviCRM is a web-based, highly customizable CRM focused on non-profit and civic sector organizations.

Its key features include case and contact management, contributions, communications, events, members, advocacy campaigns, peer-to-peer fundraisers, and reporting.

vTiger CRM Open-Source
Vtiger makes adaptable and cost-effective CRM applications best suited for small and medium businesses.

Its features include sales automation, sales enablement, project management, inventory, and other tools.

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OPEN-SOURCE PROJECT MANAGEMENT

Open-source project management aids in the management of multiple projects at the same time and staying organized.

Some top open-source project management includes:

Trello
Trello is an open-source project management software that enhances collaboration between teams. It aids the management of individual projects and ongoing workflows.

It is well suited for individuals and small teams.

Its key features include attaching files and hyperlinks, due dates, custom dropdowns, and others.

Monday
Monday is suitable for big teams to collaborate effectively, analyze progress from a single dashboard, customize their workflow and stay informed.

Some of its features include project management, email updates, progress display, and lots more.

Zoho
Zoho is an end-to-end fully customizable CRM solution for growing businesses and enterprises. It makes communication between team members seamless through its in-build chat.

Key features include email and collaboration, security and IT management, human resources, and many more.

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OPEN-SOURCE VIDEO GAMES

Many open-source video games are incorporated in Linux distributions by default and are free to use and modify, though some may be under restrictive licenses.

They can be downloaded and installed on both macOS and Windows.

Some open-source video games include:

Xonotic
Xenotic runs on Windows, macOS, and Linux. It has thrilling actions and a variety of weapons to use. It can be modified by developers to make the thrills more exciting.

OpenTTD
OpenTTD is available on Windows and Android. It allows developers to create complicated rail networks while allowing the players to find their own solutions and designs.

Battle for Wesnoth
It runs on Windows, Android, macOS, and iOS, and offers a huge variety of maps, units, classes, and campaigns using a turn-based tactical strategy.

BLOCKCHAIN OPEN-SOURCE

Blockchain open-source is a software product used to record transactions between two parties.

It documents the information from a transaction on spreadsheets accessible by all parties involved in the transaction.

Though it can not be modified, users often agree through a

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consensus to add data to the platform. It is a secured software.

It is often used in the financial sector, eCommerce businesses, online voting, e-governance, and others.

Some examples of blockchain open-source include:

Ripple
Ripple is used by businesses to manage a multitude of financial transactions. It connects corporates and banks with different service providers around the world while ensuring the security of data.

Hyperledger
Hyperledger facilitates enterprise-level business in more secured, globally conducted transactions.

Its focus is on increasing the huge array of cross-platform blockchain technology.

Ethereum
Ethereum is a software that connects businesses and users together to help them enter smart contracts. It is being continuously developed to improve its protocols in order to increase the support and user experience.

OPEN-SOURCE OFFICE SOFTWARE

Some open-source office software includes:

Abiword
AbiWord is a free word processing program. It is suitable for a wide variety of word processing tasks.

It supports other features like tables, styles, page headers

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and footers, footnotes, templates, multiple views, page columns, spell checking, and grammar checking.

OpenOffice.org
OpenOffice.org, commonly known simply as OpenOffice, is an open-source office suite.

It features a word processor, spreadsheet, presentation application, drawing application, formula editor, and database management application.

LibreOffice
LibreOffice allows users to work on text documents, spreadsheets, and databases, create presentations, edit formulas, and draw flowcharts and vector graphics.

It also allows exporting and working in different document formats.

OPEN-SOURCE ACCOUNTING SOFTWARE

Some open-source accounting software includes:

GNU cash
GNU cash is an open-source financial accounting software used for the management of financial transactions like income, expenses, etc.

It is suitable for small businesses and individuals. It is also a cross-platform application and can work on Linux, Windows, and macOS.

SQL Ledger
SQL Ledger is an accounting suite for micro, small and

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medium enterprises.

It offers basic functionality for purchasing, sales, and inventory management.

Some of its features include vouchers, a general ledger, inventory control, and many others.

Front Accounting
Front accounting is suitable for small and medium-sized companies in all industries.

Some of its key features include reporting, cash and bank management, fixed assets management, sales and accounts receivable, and many others.

Odoo
Odoo caters to businesses of all sizes in all industries.

Its key features include bank reconciliation, tax management, user management, reporting, and other additional accessories.

OPEN-SOURCE OPERATING SYSTEMS

The popular operating systems most users are conversant with, e.g. Microsoft's Windows and Apple's iOS and macOS, are all closed operating systems. However, there are a bunch of open-source operating systems out there that might interest you.

Linux Kernel
Linux Kernel offers essential functions expected of an operating system such as memory processing, data

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cancellations, and interactions with computer hardware. A lot of plug-ins have been built by developers to meet their requirements.

Linux Lite
Linux Lite is capable of running on lower-end hardware. It is a lightweight operating system designed to help users who are not conversant with Linux-based operating systems.

It has a minimal interface and is based on the Ubuntu system.

It has received regular updates over the past five years, and it includes all the capabilities, tools, and programs required of an operating system.

Linux Mint
Linux Mint is very modern and powerful. It is a Linux-based operating system, and it includes complete multimedia capabilities, which make it a user-friendly operating system.

Its distribution is Ubuntu-based, and it is more stable with better visuals than Ubuntu. It is built on the Debian platform and is very popular with beginners and experts alike.

Fedora
Fedora is widely considered to be the best open-source operating system after Ubuntu. It is Linux-based. It is built by the Fedora project community and supported by Red Hat. The community's purpose is to develop and share cutting-edge open-source technology for free.

Hence, Fedora developers focus on upstream improvements instead of creating fixes for Fedora.

React OS
React OS is a community-based operating system that may run on Windows apps. This makes it an excellent alternative

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to Windows. It has a growing community of users of about 1 million in over 100 countries. It is a developer-focused operating system.

Solus
Solus is a relatively new operating system from the Linux family, released in 2012. It currently has more than 6,000 registered users. It comes installed with LC, XChat, Transmission, Thunderbird, OpenShot Video Editor, Firefox, Budgie desktop environment, and LibreOffice suite. Its most recent version — Solus 3 was released in August 2017.

Chrome OS
Chrome OS is a part of the Chromium and Linux families with various features like security, Google cloud print, compatibility for supported android and chrome apps, virtual desktop access, cloud-based management, aura windows manager, and integrated media player. A downside of this operating system is that it only supports Nexus devices or its hardware.

Ubuntu
Ubuntu is a Linux-based operating system and is considered by most experts as one of the most efficient open-source operating systems in the world. It includes software products, such as office suite, media maps, web browser, instant messenger, etc., due to its wide adoption, it is an alternative to macOS and Windows.

Ubuntu is compatible with laptop and desktop interfaces, and is regarded as an ideal operating system for old laptops. Ubuntu includes some features like Firefox, LibreOffice, F-Spot, etc.

Xubuntu
Xubuntu is very popular with users of old XP computers. It is different from Ubuntu, and it is Unix-based. It contains

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GNOME application integrations and an inbuilt compositor. Xubuntu is known to be slow sometimes.

OpenBSD
OpenBSD is a multi-platform 4.4BSD-based Unix operating system.

It emphasizes portability, security, integrated cryptography, and standardization.

It is known to run on different hardware platforms and is considered the most secure Unix-based operating system.

FreeBSD
FreeBSD is a modern operating system used to power desktops, laptops, servers, and embedded systems with support for a large number of platforms.

Its community has been in charge of its continuous development over a thirty-year period.

Due to its advanced networking, storage, and security features, it has become a preferred platform for many busy websites and the most pervasive embedded networking and storage devices.

NetBSD
NetBSD is a Unix-like operating system that is fast, very portable, and secure.

It is compatible with a wide range of platforms, from large-scale servers and powerful desktop systems to mobile and embedded devices.

Dragonfly BSD
Like other BSD-derived systems, Dragonfly BSD is Unix-based and shares ancestor code with other BSD

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operating systems.

DragonFly has lots of useful features, chief amongst them being HAMMER. HAMMER is a modern high-performance file system with built-in mirroring and historic functionality.

DragonFly also boasts of a feature called “swapcache”, which makes use of swap space to cache file system data and meta-data.

Qubes OS
Qubes OS is a security-oriented operating system for single-user desktop computing. Qubes leverage the Xen-based virtualization, allowing the creation and management of isolated compartments called Qubes.

Some of its key features include strong isolation, template system, multiple operating systems, disposables, and many others.

Haiku
Haiku targets personal computers with its fast, user-friendly yet powerful open-source operating system. The Haiku project is made up of a single team of developers managing its kernel, drivers, tool kit, userland services, desktop applications, and preflets.

Its key distinguishing features include its focus on personal computing, unified cohesive interface, custom kernel designed for responsiveness, Rich object-oriented API for faster development, and many others.

Redox
Redox is a Unix-like operating system written in the programming language Rust, with a focus on safety, stability, and performance. It has modest compatibility with POSIX, allowing Redox to run many programs without porting.

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It was built to run most Linux programs with minimal modifications, hence, it is an alternative to Linux on computers.

AROS

The AROS Research operating system is a desktop operating system designed to be lightweight, flexible, and efficient. It's an independent, portable and free project, aiming at being compatible with AmigaOS at the API level (like Wine, unlike the UAE), while improving on it in many areas.

It was designed to run as a stand-alone 'native' version, bootable directly from a hard disk — or hosted, opening a window on an existing OS — to develop software and run Amiga and native applications at the same time. It can also be easily ported to different kinds of hardware architecture and processors, such as x86, PowerPC, Alpha, Sparc, HPPA.

FreeDOS

FreeDOS is a complete, free, DOS-compatible operating system that you can use to play classic DOS games, run legacy business software, or develop embedded systems.

While FreeDOS provides some utilities, it runs any program intended for MS-DOS. For example, Windows standard-mode works on FreeDOS, but ‘386-mode Windows for Workgroups 3.11 does not.

BASIC-DOS

BASIC-DOS is known to support a FAT file system, COM and EXE executable formats, and many of the same PC DOS APIs, commands, and data structures, including Program Segment Prefixes (PSPs) and File Control Blocks (FCBs). Nonetheless, BASIC-DOS does not pursue compatibility as its primary goal. Its goal is to show what could have been achieved as the first IBM PC operating system, not to create a “clone” of PC DOS and continuously chase compatibility problems.

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skiftOS
SkiftOS is a hobby operating system built from scratch in modern C++ for learning and for fun, targeting x86 platforms. Its features include a kernel named "hjert", a reactive UI library, a graphical interface with a compositing window manager, a strong emphasis on user experience, and familiar Unix utilities.

Tock
Tock is an embedded operating system designed for running multiple concurrent, mutually distrustful applications on low-memory and low-power microcontrollers.

Its core features include safety, reliability, and seamless low-power.

Its architecture provides isolation in the kernel using a language sandbox. The kernel components comprising drivers and virtualization layers are isolated from each other with no resource overhead.

Essence
Although still in development, Essence is a desktop operating system built from scratch, for control and simplicity.

Essence has at its core: efficiency and performance. It can run on low-powered hardware on less than 30 MB of drive space and boot with even less RAM. It starts up in seconds and shuts down instantly, even on old hard drives.

Subgraph OS
Subgraph OS is a modern desktop computing and communications platform that is designed to be unaffected by network-borne exploits and malware attacks. Its features include strong system-wide attack mitigations that protect its core operating system and all applications. Key applications are run in sandbox environments to reduce the impact of any

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attacks.

It was designed to promote communication and collaboration without fear of surveillance or interference by sophisticated foes through network-borne attacks. Its design reduces risks in endpoint systems so as to make attacks difficult.

Osv.io
Osv.io is a versatile modular unikernel designed to run unmodified Linux applications securely on micro-VMs in the cloud. Its key features include superior performance, zero OS management, rapid VM build and deploy, optimized native app, and many more.

Barrelfish
Barrelfish is a research operating system being built collaboratively from scratch by ETH Zurich and Microsoft Research and is now partly supported by HP Enterprise Labs, Huawei, Cisco, Oracle, and VMware.

It was designed as a vehicle prospecting ideas about the structure of operating systems for the hardware of the future. It runs on x86 platforms, with ports to other platforms still in progress.

ToaruOS
ToaruOS is an educational resource that provides a representative microcosm of functionality found in major desktop operating systems. It comprises its own composited windowing system, kernel, advanced code editor, C standard library, dynamic bytecode-compiled programming language, bootloader, and other utilities.

It runs on x86-64 PCs and offers experimental support for ARMv8.

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Mezzano
Mezzano is an operating system written in Common Lisp, except for some assembly code. Its demo releases are designed to run in VirtualBox with 2 GB of RAM, a virtio-net NIC, and an Intel HDA audio controller.

House
House is written in Haskell as a demo software and running in a standalone environment. It is a platform for exploring multiple ideas on low-level and system-level programming in a high-level functional language.

Minoca OS
Minoca OS is a modular, lean, and flexible general-purpose operating system designed for feature-rich embedded devices. It offers all of the high-level functionality of an operating system, but at a fraction of the usual memory requirement.

It is designed with a unified power management architecture that allows for smarter power management decisions, leading to better device battery life. It has a minimal OS footprint and boots to a shell in 5 MB of RAM.

Kerla
Kerla is a monolithic operating system kernel with Linux binary compatibility, written from scratch in Rust. It runs Linux binaries without any modifications.

It supports QEMU and Firecracker (with virtio-net device driver), and runs on x86_64 devices. It has a Docker-based initramfs build system.

SOS
SOS was written for fun as a tiny toy operating system in Rust. It was inspired by a series of blog posts and “how-to” on operating systems.

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PowerNex
PowerNex is made up of the words — Power and Next — as an indication of its intent to Power the Next generation of hardware.

It is an operating system written in the D programming language. Its goal is to have an operating system written in D, where the PowerNex kernel powers its core.

It requires a 64-bit processor and a minimum of 512 MB of RAM.

Interim OS
Interim OS is an operating system created based on minimalism. It borrows its concept from Lisp machines and Plan 9. It boots to a JITting Lisp-like REPL and offers the programmer/user the system's resources as file systems.

KolibriOS
KolibriOS is an operating system that boots in less than a few seconds from power-on to working GUI. It is a small but powerful and fast operating system that requires 8 MB of RAM and a few megabytes of disk space.

Its features include a word processor, graphical support, web browser, image viewer, and lots of games.

MINIX
MINIX is an operating system focused on achieving high reliability through fault tolerance and self-healing techniques. It can be used as a base for research projects or for embedded systems.

It is a microkernel that runs in kernel mode. Its design ensures that failures of the system due to bugs or attacks are isolated.

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RISC OS
RISC OS is a small but fast full desktop operating system. Its core system and a few apps fit inside 6 MB. It is not based on any other OS, rather it is the first ARM OS, begun in 1987 by the team who designed the original ARM processor.

It is fast and responsive on modern hardware. Typically, the memory consumed by apps is usually counted in kilobytes.

HelenOS
HelenOS is a portable microkernel-based multiserver operating system, designed and implemented from scratch. It decomposes key operating system functionality into a collection of fine-grained user space components that interact with each other via message passing.

Has a mix of clean design and coding style together with portability and modularity.

GeckOS/A65 V2.0
GeckOS/A65 V2.0 is a multitasking and multithreading operating system.

Its features include multithreading, dynamic memory management, relocatable file format, lib6502 standard library, and internet support.

Serenity
Serenity is a graphical Unix-like operating system for desktop computers. It is a merger between the aesthetic productivity software of the 1990s and the power user accessibility of the 2000s.

It is described as a love letter to '90s user interfaces with a custom Unix-like core.

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GNU Hurd
The Hurd is the GNU project's replacement for the Unix kernel. The Hurd is a collection of protocols formalizing how different components may interact. Its servers/cloud run on the Mach microkernel to implement file systems, network protocols, file access control, and other features that are implemented by the Unix kernel or similar kernels (such as Linux).

Plan 9
Plan 9 is a distributed operating system built on Unix concepts. It is built on a per-process namespace and a simple message-oriented file system protocol. Although it shares numerous command-line utilities with Unix counterparts, it works differently. Its distributed operation relies on the per-process namespaces as well, allowing client and server processes to communicate across machines.

Harvey
Harvey is an experimental operating system based on the MIT-licensed release of Plan 9.

The goals of its development include:

- The modernization of existing C code;
- Improvement of the kernel to allow the inclusion of kernel code implemented in new languages such as Rust;
- To ship a useful Go-based user space;
- To build an easily deployable Plan 9 distribution.

Illumos
Illumos is a Unix operating system providing next-generation features for downstream distributions. These features

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include: Virtualization options, a next-generation file system, downstream distributions, networking, and advanced system debugging.

It is superb for both traditional and cloud-native deployments.

AquilaOS
AquilaOS is a complete Operating System (kernel + system) that is designed to be POSIX compliant and mostly ISA transparent. It is user-friendly and has all its implemented interfaces compliant with POSIX.

It supports multitasking and multithreading, sessions, process groups, and job control, virtual file system (vfs), devices subsystem (kdev) using major/minor numbers, and memory management subsystem (with COW and allocate on demand).

Sculpt
Sculpt is a general-purpose operating system that combines Genode's microkernel architecture, capability-based security, sandboxed device drivers, and virtual machines in a novel operating system for commodity PC hardware.

Sculpt is used as a day-to-day OS by the Genode developers.

Biscuit
Biscuit is a monolithic, POSIX-subset operating system kernel in Go for x86-64 CPUs. Its aim is to study the performance trade-offs of using a high-level language with garbage collection to implement a kernel with a common style of architecture.

Its features include:

- Multicore;
- Kernel-supported threads;

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- Journaled FS with concurrent, deferred, and group commit;
- Virtual memory for copy-on-write and lazily mapped anonymous/file pages;
- TCP/IP stack;
- AHCI SATA disk driver;
- Intel 10 GB NIC driver.

Xv6

Xv6 was developed as a teaching operating system for MIT's operating system course. It was created to teach operating systems engineering.

Open Vs. Closed Source Operating System

- The price of open-source software OS is often low or zero, while the price of closed source OS is more expensive, with the costs already factored into the prices of machines;
- Open-source OS offers little or no restrictions based on usability and modification of software, while closed source OS has lots of restrictions on users based on usability and modification of software;
- The source code of open-source OS is available to the public, hence, vulnerabilities can be quickly detected and fixed, while the source code for closed source OS is accessible only by the owners or companies, hence vulnerabilities might go unnoticed for a longer period of time;
- Examples of open-source OS include: Linux and Ubuntu;

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- Examples of closed source OS include: Windows and iOS.

Advantages and Disadvantages of Open-Source Operating System
Advantages of open-source OS include:

- It is reliable and efficient as a result of multiple developers working on it simultaneously;
- They are mostly free or cost very little in price;
- They can be easily customized to fit any requirement.

Disadvantages of open-source OD include:

- It is not user-friendly. Some Measure of technical knowledge is usually required to use them;
- It poses a security risk because many people have access to its source code;
- No readily available customer support service to assist users.

OPEN-SOURCE MOBILE OPERATING SOFTWARE

Open-source operating system software is also popular on mobile. Let us take a look at some of them below:

Android
Android is an open-source operating system for mobile devices and a corresponding open-source project led by Google. But, due to the amount of proprietary software that

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accompanies Android on consumer devices, many users still do not consider it an open-source operating system. Android's aim is to avoid a central point of failure that will empower any individual or group to restrict or control the innovations of any other player. Consequently, Android is a full, production-quality operating system for consumer products, complete with customizable source code that can be ported to nearly any device and public documentation that is available to everyone.

Plasma Mobile
Plasma Mobile is an open-source user interface and ecosystem for mobile devices, built on the KDE Plasma stack. It runs on a Linux distribution.

Though Plasma mobile applications are developed with Kirigami, applications developed for other Linux operating systems are generally compatible.

Plasma mobile aim is to contribute to open standards and implement and develop the same in a transparent process accessible to all.

postmarketOS
PostmarketOS is a touch-optimized, pre-configured Alpine Linux with its own packages, which can be installed on smartphones. The biggest upside is that Alpine is small. The base installation is about 5 MB.

Since Android and iOS are known to stop providing updates after a few years, the postmarketOS aim is to enable a 10-year life cycle. PostmarketOS is still in the early phases of development.

PureOS/Librem
In a bid to achieve convergence, the security and privacy-focused company Purism built its own PureOS Linux

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distribution. They are also developing a Linux-based, secure smartphone called Librem 5. PureOS is a fully-convergent, user-friendly, secure, and freedom-respecting operating system for your daily usage.

Convergence means having the same operating system running on desktops, tablets, and smartphones.

It is believed that Librem smartphones will run on PureOS. In addition to PureOS, it can run GNOME and KDE, with Debian, Ubuntu, SUSE, Fedora, and even Arch Linux.

Ubuntu Touch by UBports

Ubuntu Touch by UBports was an attempt by the king of the desktop Linux world — Ubuntu, to extend its territory with an open-source mobile operating system. In a bid to achieve convergence, the parent company of Ubuntu — Canonical, teamed up with some device manufacturers to launch Ubuntu Phone. The first few devices were targeted at developers.

Though unsuccessful after a few years, UBports still keeps Ubuntu Touch alive. Ubuntu Touch's sole aim is to provide the best and most secure user experience; better than traditional systems, where these concerns are balanced against profit.

Tizen

Tizen is a Linux-based mobile operating system built from the ground up to address the needs of all stakeholders of the mobile and connected device ecosystem. The Tizen project is supported by the Linux foundation, hence, it is often referred to as an official Linux mobile operating system.

Albeit its failure, the first Tizen phone — Samsung Z was released in the South Asian market. It could not survive in the competitive market.

Mobian OS

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The Mobian OS aim is to bring Debian GNU/Linux to mobile devices. Currently, it only supports PinePhone. Mobian integrates the standard Debian distribution with the Phone Shell (phosh) project in a package that works on certain mobile phones like the Pinephone. Subsequently, the idea is to minimize the Mobian-specific pieces by “upstreaming” customization to the original projects.

LineageOS
LineageOS is a free and open-source operating system for various devices, based on the Android mobile platform. LineageOS promises to push for user personalization and preference. It extends the functionality and lifespan of mobile devices from more than 20 different manufacturers.

/e/
/e/ previously known as "eelo" is an open-source mobile operating system with its own associated web services like cloud storage, email, office suite, and many more.

Its aim is to improve the privacy and security of users' data, as well as empower end-users with all tools of a standard operating system.

Named after the eel fish and its ability to hide in the sea, indicative of privacy, /e/ is forked from LineageOS, an open-source Android distribution.

GrapheneOS
GrapheneOS is a mobile operating system focused on privacy and security technology, compatible with Android Apps.

It is essentially a hardened version of Android without Google services. Its research and development cover substantial improvements to sandboxing, exploit mitigations, and the permission model.

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GrapheneOS also includes: Externally developed Seed Vault encrypted backup, security-focused PDF viewer, modern privacy/security focused-camera app, and Auditor app/attestation service providing local, and remote verification of devices

CalyxOS
CalyxOS is an Android-based mobile operating system with a heavy focus on security and privacy. Though it currently supports only a few devices, it is regarded as one of the best options as a Google-free operating system with regular updates. Furthermore, Lineage uses some of CalyxOS apps for some functions as a backup.

Its design is targeted to meet the extraordinary privacy and security needs of exposed persons like activists, journalists, human rights defenders, and lots more.

WEB SERVERS

A Web server is a computer system that utilizes Hypertext Transfer Protocol (HTTP) to respond to clients' requests by distributing data from web pages to end-users through the internet. Some open-source web servers are below.

Apache HTTP Server
Apache HTTP Server is an open-source, powerful, and flexible HTTP/1.1 compliant web server. It was designed for modern operating systems like Windows and Unix. According to W3Techs, the Apache HTTP Server is the most popular web server, with 45% of websites using it.

It was launched in 1995 with the goal of providing an efficient, secured, and extensible server that provides HTTP services in sync with the current HTTP standards.

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NGINX
NGINX is an HTTP and reverses proxy server, a mail proxy server, and a generic TCP/UDP proxy server. It is the second most popular server.

Some of its basic HTTP server features includes: Accelerated reverse proxying with caching, load balancing and fault tolerance, Modular architecture, SSL and TLS SNI support, and others.

Node.js
Node.js is a server-side cross-platform JavaScript environment designed for building and running network applications like web servers. It is an asynchronous event-driven JavaScript runtime design to build scalable network applications. It was built on Chrome's V8 JavaScript engine. Despite being designed without threads, you may still take advantage of multiple cores in your environment.

Apache Tomcat
Apache Tomcat (called "Tomcat" for short) is an open-source implementation of the Jakarta Servlet, Jakarta Expression Language, and WebSocket technologies. It is used to run Java applications based on the provision of a ‘pure Java’ HTTP web server environment. It was released under the Apache License 2.0 version.

Lighttpd
Lighttpd is a web server that is fast, secured, compliant, and optimized for high-performance environments. It has lower resource use in comparison to other popular servers. The use of resources such as memory and CPO is efficient.

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DATABASE SYSTEMS

Database Systems is a software that collects electronic and digital records to extract useful information and store that information. Below, there are some open-source database systems:

MySQL

MySQL is a new generation transactional data store. It is one of the most widely used relational embedded databases. MySQL is the most trusted and widely used open-source database platform in use today. Some key features of the latest version of MySQL 8.0 include:

- MySQL Document Store
- Transactional Data Dictionary
- SQL Roles
- Default to utf8mb4
- Common Table Expressions
- Window Functions
- And others

PostgreSQL

PostSQL is an object-relational database system that utilizes and extends the SQL language together with features that ensure the safe storage and scaling of complicated data workloads. It supports macOS Server, most distributions of Linux, and Microsoft Windows.

PostgreSQL is well known for its proven architecture, reliability, data integrity, robust feature set, and extensibility.

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MariaDB

MariaDB is a distributed SQL database solution. It supports transactional, analytical, and combination workloads cum relational, JSON, and hybrid data models.

It can deploy standalone databases and data warehouses to fully distributed SQL, which executes millions of transactions per second and performs interactive, ad hoc analytics on billions of rows.

Since it is a fork of SQL, its interface for accessing data is similar to SQL.

Apache Hive

Apache Hive facilitates the reading, writing, and management of large datasets residing in distributed storage using SQL. It can also be used for writing SQL-like queries to pull data from the Hadoop distributed file system (HDFS) and other compatible systems. Apache Hive can be used as both a database and a data warehouse.

SQLite

SQLite is a transactional database engine that is self-contained, and serverless, with zero configuration. It is one of the most used database engines in the world. It is built into all mobile phones and most computers, and comes bundled inside numerous other applications that are used daily. The SQLite file format is cross-platform, stable, and backward compatible.

MOBILE DEVELOPMENT FRAMEWORKS

Ionic

Ionic is an open-source UI Software Development Kit (SDK)

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for building modern high-performance cross-platform apps from a single codebase. In addition to native apps, Ionic allows the building of progressive web apps.

React Native
React Native is a framework for building fast native apps with JavaScript and react. Codes can be written with JavaScript and shared across platforms, while computational heavy features can be implemented with native modules for iOS and Android. React Native provides access to native mobile controllers, which allows it to achieve a native look and feel to apps.

Flutter
Flutter is a multi-platform Software Development Kit (SDK) from Google for creating mobile, desktop, and web apps with a single codebase. Since it is a full-fledged SDK (it is not a framework or library), it includes ready-made widgets, rendering games, testing and integration APIs, etc., users of Flutter must know the Dart programming language.

Xamarin
Xamarin is a cross-platform mobile app development tool owned by Microsoft, which lets the engineers share up to 90 percent of the code across major platforms. Xamarin uses the C# programming language and is based on the .NET framework. Although initially started as a commercial project, it became a popular cross-platform product for developing mobile apps within the Microsoft ecosystem after its acquisition by Microsoft.

QA AUTOMATION TOOLS

Selenium
Selenium is a suite of testing automation tools, and it is open-source. It is often referred to as the de facto test

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automation tool in the quality assurance world. It supports mobile testing, browsers, operational systems, and numerous programming languages. It is used in production by companies like Google, Fitbit, Netflix, HubSpot, etc., it supports all operating systems and browsers.

Watir
Watir (Web Application Testing In Ruby) is an open-source Ruby library for automating tests. It is used for executing simple automated tests. It interacts with browsers the same way users do, i.e. by clicking inks, validating texts, and filling forms. It is supported on Chrome, Safari, Firefox, Internet Explorer, and Edge.

Robot Framework
Robot Framework is a generic open-source automation framework for keyword-driven testing. It is used for test automation and robotic process automation.

It can be integrated with other tools to create powerful and flexible automation solutions. Robot Framework has an easy syntax, using human-readable words. It has a rich ecosystem that consists of libraries and tools developed as separate projects.

BIG DATA ANALYTICS TOOLS

Apache Hadoop
The Apache Hadoop software library is the most popular framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models.

Its design ensures it is capable of scaling up from single servers to thousands of machines, each offering local computation and storage. The library is responsible for the

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detection and handling of failure at the application layer, thereby delivering a highly-available service on top of a cluster of computers, each of which may be prone to failures.

Apache SAMOA
Apache SAMOA (Scalable Advanced Massive Online Analysis) is an open-source platform that analyzes streaming data and develops ML algorithms without using distributed stream processing engines like Apache Storm or Apache Samza.

HPCC
HPCC (High-Performance Computing Cluster) is a proven solution for big data analytics that can be implemented by various businesses. HPCC enables businesses to better analyze and comprehend data at scale, improving business time to results and decisions. HPCC systems provide a consistent data-biased programming language, two processing platforms, and a single, complete end-to-end architecture for efficient processing.

OPEN-SOURCE WEBSITE BUILDERS

Website builders essentially allow users to build a website without coding. Many of them offer an intuitive user experience so that users would not spend too much time building their websites. Some open-source website builders are below.

Droopler
Droopler is a Drupal CMS (Content Management System) based on a Drupal distribution website builder. It is a pre-built complete website, easy to customize and manage. It allows users to build their websites from ready-made components, making corn unnecessary.

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Droopler is SEO optimized, editor-friendly, and extendable.

Drupal
Drupal is a content management software used for building websites without using codes.

Some features of Drupal include: Reliable performance, excellent security, and easy content authoring. Drupal has modularity as one of its core principles. It offers themes, which lets users customize their content's presentation.

Joomla
Joomla is a website builder that lets you build, manage and publish websites. It comes with several default templates. These templates allow you to modify features like fonts, colors, and layouts without coding.

Microweber.org
Microweber.org builds websites using the laravel framework. It contains a default layer called Dream. Some features of microweber.org include: Dynamic layouts, blog features, an online store builder, and statistics.

WordPress
WordPress (wordpress.org) is an open-source website builder, which allows you to create and edit pages and blog posts. It is different from wordpress.com. It offers three default themes: Preformatted layouts you can use without coding, and over eight thousand themes available on the WordPress webpage. Some of its features include: Plugins, media management, and SEO.

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FREE AND OPEN-SOURCE CYBERSECURITY TOOLS

Kali Linux
Kali Linux is an open-source Debian-based Linux distribution for various information security tasks, and offers cybersecurity utilities and penetration testing tools. It comes with pre-packaged tools for reconnaissance and delivering payloads, together with other penetration-testing tools.

Kali Linux uses Windows Subsystem for Linux (WSL), which allows the running of Linux executable files directly from a Windows 10 system.

KeePass
KeePass is an open-source password manager that helps you to manage all your passwords in a secure manner. It enables users to have a single database for their unique passwords, network login credentials, websites or email accounts, which is then locked with one single master key to unlock the whole database. It uses the most secure encryption algorithms available.

Metasploit Framework
Metasploit Framework is a tool used for exploitation and vulnerability validation, which you can use to offensively test your systems for known and open vulnerabilities.

It can also be used for auditing and network port scanning.

Metasploit Framework has an auto-exploitation feature that works by cross-referencing open services, vulnerability references, and fingerprints to find corresponding exploits.

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Nikto
Nikito is a web server scanner capable of performing comprehensive tests for multiple vulnerabilities. Its tests include over 6700 potentially dangerous files/programs, checks for outdated versions of over 1250 servers, and version-specific problems on over 270 servers.

Users can also check for server configuration items such as the presence of multiple index files, and HTTP server options.

Nmap
Nmap (Network Mapper) is a tool used for penetration testing and security auditing. It maps networks and ports before a security audit is started, and then uses NSE scripts to detect recognizable security problems. The script detects vulnerabilities, misconfigurations, and security issues concerning network services.

It is designed for scanning large networks, and it runs on Limus, Windows, and macOS.

OpenVAS
OpenVAS is a full-featured vulnerability scanner. Users can carry out unauthenticated and authenticated testing, performance tuning for large-scale scans, various high-level and low-level internet and industrial protocols, and a powerful internal programming language to implement any type of vulnerability test.

OSSEC
OSSEC is a scalable and multi-platform Host-based Intrusion Detection System (HIDS). It is capable of resisting cyber-attacks and system changes in real-time utilizing firewall policies, integration with third parties such as CDNs, and support portals. It had a self-healing capability.

Security Onion

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Security Onion is a scalable platform and a Debian-based Linux distribution for identifying threats, enterprise security monitoring, and log management. It includes security tools like NetworkMiner, OSSEC, Zeek, Logstash,Kibana, Snort, Squert, and many others for protection against cyber threats.

VeraCrypt

VeraCrypt is a disk encryption software that runs on Windows, macOS, and Linux. It encrypts an entire partition or storage device before dumping it in the cloud. It encrypts in real-time. It also allows users to pre-boot authentication by encrypting a partition or drive where Windows OS is installed. It also supports hidden files and hidden operating systems on a machine.

Wireshark

Wireshark is a widely used network protocol analyzer. It enables users to observe networks at a microscopic level by viewing the traffic, dumping of specific packets, checking the packet format, and finding network issues this way. It offers live capture and offline analysis of data. It can also be used for decrypting multiple protocols.

OPEN-SOURCE AUDIOVISUAL PRODUCTION TOOLS

There is plenty of open-source software used as audiovisual production tools. Many of them can be used in graphics, audio, video, and 2D and 3D animations.

Graphics

The tools below can be used to create graphics like banners, logos, print, etc.

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Inkscape
Inkscape is an open-source vector graphics editor used to design SVG and PDF files in the RGB color. It is very useful for manipulating SVG maps and charts for web applications. It allows users to open files with the integrated XML editor and see all the parameters of an object.

GIMP
GIMP is an image editor capable of manipulating color, cropping and resizing images, and file optimization for the web. It can run on macOS, Windows, and many more operating systems.

Krita
Krita is a cross-platform application offering an end-to-end solution for designing digital art files. It supports many fields of paintings like concept art, matte, painting, illustrations, animations, and textures.

Scribus
Scribus is one of the premier open-source desktop publishing applications. It can be used to create a complete document; convert PDF from Inkscape; or LibreOffice from RGB to CMYK. Scribus supports professional publishing features, such as CMYK colors, spot colors, ICC color management, and versatile PDF creation.

RawTherapee
RawTherapee is a cross-platform raw photo processing system. It can develop raw files from digital cameras. It provides a suite of tools to produce professional photos. It offers a very powerful engine and a non-destructive editor.

LibreOffice Draw
LibreOffice Draw is a professional desktop publishing solution. With LibreOffice Draw, you can produce anything from a quick sketch to a complex plan, and it gives you the

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means to communicate with graphics and diagrams.

It allows users to manipulate pictures and graphical objects, and produce technical drawings, brochures, posters, and many other documents.

Audio

Ardour

Ardour is an open-source digital audio recording system for recording and editing audio. It is flexible and supports plugins. It allows users to edit, listen and compare modifications.

Audacity

Audacity is an open-source audio software that features a multi-track audio editor and recorder for Windows, macOS, GNU/Linux, and other operating systems.

LMMS

LMMS is a music production tool. It supports plugins, and you can play drum samples with a step sequencer, edit instruments, and a host of other things.

Mixxx

Mixxx allows users like DJs to integrate the tools they need to perform creative live mixes with digital music files. It supports MIDI controllers, time-coded discs, and dedicated sound cards. You can manage your music library and add effects.

Video

VLC

VLC is a cross-platform multimedia player and framework known for playing most multimedia files including DVDs, Audio CDs, VCDs, and various streaming protocols. It can also be used to convert a video into another codec or container or to recover a broken video.

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OpenShot
OpenShot is a video editor which is easy to learn and use, with popular features and capabilities. Its features include animations and keyframes, audio waveforms, 3D animations, Slow Motion & Time Effects, and many others.

Shotcut
Shotcut is an open-source, cross-platform video editor. It supports 4K and professional codecs.

Blender Velvets
Blender is a video editing, compositing, and effects tool famous for 3D animation. It is capable of using add-ons and transforming them into a complete suite for 2D video editing. It comprises extensions and scripts, created by moviemakers, that transform the Blender 3D creation software into a 2D video editor.

Natron
Natron is a node-based compositing software. It is an alternative to Adobe's After Effects, but works differently. Its programming ensures that it uses a set of techniques with the aim of composing complex images through the arrangement and organization of various sources like videos, photographs, static and moving images, etc.

OBS
OBS (Open Broadcaster Software) is used for video recording and livestreaming. It is a leading solution for recording or livestreaming e-sports and video games on YouTube or Twitch. It can also be used to record users' screens.

2D and 3D Animation

Blender
Blender is a 3D creation suite. You can do an entire project directly in Blender, or use it to create 3D effects for a movie or video as it supports the entirety of the 3D pipeline —

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modeling, rigging, animation, simulation, rendering, compositing, and motion tracking, even video editing, and game creation.

Synfig Studio
Synfig Studio is an open-source 2D animation software. It is designed as a solution for creating film-quality animation using a vector and bitmap artwork. With Synfig studio, users do not need to create animation frame-by-frame, as it allows users to produce 2D animation of a higher quality with fewer people and resources.

TupiTube
TupiTube is a 2D animation tool. It is capable of transforming drawings or pictures into a video or creating an animated GIF and small loops.

OPEN-SOURCE DEVELOPMENT TOOLS

VS Code
VS Code is a lightweight but powerful, streamlined code editor. It runs on Windows, macOS and Linux. It features built-in support for TypeScript, Node.js, and JavaScript. It supports development operations like debugging, task running, and version control. It has a rich ecosystem of extensions for other languages (such as C++, C#, Java, Python, PHP, Go)

Budibase
Budibase is an open-source low-code platform for creating internal apps, workflows, and admin panels in short times. With Budibase, data and processes can be transformed into powerful internal tools within a few minutes. It supports PostgreSQL, MySQL, MSSQL, MongoDB, Rest API, Docker,

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K8s, and others.

Vercel
Vercel is a platform for static sites and frontend frameworks. It provides frontend developers the ability to host websites and web services that deploy instantly and scale automatically – all without any configuration.

Oh My Zsh
Oh My Zsh is an open-source, community-driven framework for managing your Zsh configuration. It features over 275 plugins for developers.

Gitlab
GitLab is a DevOps software that offers users the ability to develop, secure, and deploy software in a single application. It is a single application that spans the entire software development lifecycle. Its functionality allows collaborative planning, building, securing and deployment as a complete DevOps platform.

Supabase
Supabase is an open-source Firebase alternative. It is a suite of open-source tools, joined together to produce a seamless developer experience. Its features include authentication, storage, edge functions, database, etc.

PostHog
PostHog is an open-source product analytic platform for product data, with a data warehouse and an API. It comes with tools (like session recording, heatmaps, and feature flags) to improve users' products.

Snyk
Snyk (pronounced sneak) is a developer-first security platform for building cloud-native applications, securing code, dependencies, containers, and infrastructure as code. Snyk

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provides auto-remediation of security vulnerabilities, direct integration into your software development life cycle, visualization of dependencies, and support for all the major IDEs.

Prisma
Prisma is a next-generation ORM consisting of the following parts:

- Prisma Client: Auto-generated and type-safe query builder for Node.js & TypeScript;
- Prisma Migrate: Migration system;
- Prisma Studio: GUI to view and edit data in your database.

Prisma offers a type-safe API that can be used across the frontend and backend.

Storybook
Storybook is an open-source tool for UI development. It provides a powerful environment that allows teams to design, create and organize UI components. It integrates well with popular front-end frameworks like Vue, React, and Svelte.

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HOW OPEN-SOURCE SOFTWARE COMPANIES MAKE MONEY

Since the source code of open-source software companies is available for anyone to access for free, why would anyone pay for it? How do these companies make money?

Well, the use of source-code is guided by the terms and conditions contained in the license. There are numerous license schemes, some with restrictions such that the use of a source code may not be available for free for for-profit organizations. Many open-source companies have grown in leaps and bounds into multi-billion dollar enterprises with the right combination of revenue-generating models.

DONATIONS

Some platforms like GitHub, Patreon, and Buy Me A Coffee allow interested individuals and organizations to make donations to support open-source projects that interest them. Individual developers and the community of developers make a substantial amount of money from such donations.

A disadvantage of this is that it is neither a predictable, nor sustainable means of income.

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HOSTED VERSION OF THE PRODUCT

You could obtain an open-source software and run it on your own server without making any payment to the open-source company. If you want the open-source company to host it on their own server, then you would have to pay for a hosted version of the product. Paying for a hosted version of an open-source software saves you the worries about backups, downtime, upgrades, etc.

The challenge with this model is that margins could be too low. Too high a margin will make users justify the cost of maintenance of their own servers, hence, refusing to pay for a hosted version of the software.

PAID SUPPORT OR COURSES

In order to deploy and maintain the software, payment is made to the open-source company to carry out all of these activities on behalf of the user (often a commercial company).

Red Hat uses this model to finance its open-source software through payment for support contracts and on-site configuration. Commercial companies weighing the cost of in-house troubleshooting vs contracting it out to the software owners, companies might prefer to contract it out to the software owners instead of doing it in-house as it might be too expensive for them.

The downside of this is that over time, companies that rely heavily on the open-source product will naturally invest in their own engineering efforts to understand the product, thereby reducing the need for external support. Consequently, there is a small usage window where this model works.

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In order to make their software more accessible, open-source companies can also offer training courses at a lower cost than hands-on management and support.

OPEN CORE

Open-source companies could make the majority of the codebase to be open-source, while a smaller part is made proprietary. Usually, the small proprietary part is targeted at commercial users. Some open-source companies sometimes offer extra features on their products that commercial customers need to use the product at a large scale. Such features might include advanced user management and specialized integrations.

The challenge with this model is the ability to maintain a balance between what is to be given free versus what is to be kept proprietary. If too much is given for free, the company loses money while if too little is given for free, then the idea of open-source is defeated.

DUAL LICENSING

Dual Licensing is quite similar to the open core model. Some open-source companies provide dual licenses. One of the licenses might allow a small, independent developer to use the software for free, while the other license will require payment from for-profit companies.

SOFTWARE AS A SERVICE

The Software as a Service (SaaS) model is most viable for open-source projects that have witnessed plenty of demand, and offer complete applications like marketing automation tools. Open-source companies host the project themselves

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so that users can focus on using the tool instead of being distracted by security and maintenance if hosted by the user. Hence, users pay a regular fee to use the hosted solution. The fees are typically monthly or annually. Furthermore, it is a great solution for users who lack the technical expertise to host the solution themselves. SaaS is a popular model of making money for open-source companies.

PAID FEATURE REQUESTS

Sometimes, developers can find a commercial company interested in their project or one already using their project; and offer to develop new features based on the company's unique needs. In return, the company pays for the added features. Oftentimes, this is cheaper for the company than to have their own developers trying to figure out the codebase and add new functionality.

GITHUB SPONSORS

In order to support the work of developers, GitHub introduced GitHub sponsors as a means via which developers can receive funding from the community that depends on their work, seamlessly through their GitHub profiles. This was introduced on May 23, 2019.

GitHub Sponsors charge zero fees. 100% of sponsorships go to the developers. It aims to reward developers for maintaining free software.

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GET PAID TO BUILD OPEN-SOURCE EXTENSIONS FOR EXISTING PRODUCTS

Another way for developers to make money from existing products is by helping commercial companies develop a plugin or connector that integrates a product with a different tool.

Sometimes, a company may lack the SDK needed to integrate a product into their project. In this case, a developer could also help develop an SDK for a programming language that wraps the product's API.

SELLING OTHER PRODUCTS

In addition to the open-source projects, some open-source companies might choose to generate revenue from a suite of related or unrelated proprietary products. It is easy to manage if their products support their open-source ones. The challenge of building and maintaining multiple products could be tedious for small teams, hence, the open-source product could suffer.

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WHAT EVERY OPEN-SOURCE COMPANY NEEDS BEFORE IT CAN CONSIDER MAKING MONEY

Before any open-source company starts making money, there are two prerequisites needed.

1. Broad Adoption: By the nature of the open-source idea, open-source companies give most of their developed software away for free. They are only able to capture a small amount of created value from their community of users. Since most users are able to get most of their needs solved by zero-charge open-source products, many will never pay for the software. Hence, the conversion rate of users to paying customers is usually low. Therefore, for an open-source company to capture a substantial amount of value, there has to be broad adoption of the product, i.e. the open-source project needs to have a large user base and community. With a large community of users, that small conversion rate can raise a substantial amount of revenue for the company;

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2. Primary Credibility: Given the low monetization rates, the second prerequisite is primary credibility within the community. This is necessary because it affords the open-source company to drive efficiency in the sales and marketing process. Possessing primary credibility implies that the open-source company is the go-to place for assistance with the software. Lacking this, the company might be in stiff competition with others in the market for that attention.

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WHEN TO USE OPEN-SOURCE FOR YOUR PRICING MODEL

There is seldom a one-size-fits-all solution to the question of the open-source pricing model. It is a reality in the world of business that you have to do your own research, brainstorm, consider lots of alternatives, and then reach your own conclusion.

Nonetheless, there are a few places where open-source is a great option:

- Software infrastructure: If your software is going to be a critical asset in your customers'' software stack, then open-source is a great option for you. Users do not like getting bogged down trying to decipher proprietary software or having downtime due to issues with your PaaS. If users can self-host the software, it increases the adoption rate;
- Developer Tooling: If your target market is developers, you are better off being open-source. Developers are a bunch of curious beings who love to build their own solution if a solution is proprietary, or sometimes due to debugging requirements for production software;
- Sensitive Data: Due to a massive trust deficit amongst people, it takes a whole lot of trust to share data. Hence, if you're building a product that involves

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sensitive data, you are better off using open-source to help earn trust;

- Security or Compliance: If your company or product is within the security or compliance industry, open-source might be a good fit for you. You will be able to build a more secure product if people are able to easily vet it. Self-hosting by customers also makes sure that you are able to easily comply with their requirements;

- You offer a free-tier: A free-tier product might require an open-source for the functionality of the free-tier. You may choose to make higher tiers proprietary, especially for large customers who might need more customized solutions.

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WHY YOU NEED AN OPEN-SOURCE SOFTWARE STRATEGY

Does your company use a substantial amount of open-source products? How involved is your company in steering the open-source project it uses heavily? Would outside contributors benefit from an in-house project you have started?

These questions all point to the fact that this is the age when companies need to make a conscious effort to design an open-source software strategy.

Below, we shall look at some reasons you need an open-source software strategy:

1. Your budget does not cover your increasing software maintenance cost. You need the money to cover other critical aspects of your business;

2. Reduce dependency on proprietary software. Do not allow yourself to be dragged through the rigors of never-ending software upgrades, costing you a ton of money;

3. Open-source software allows you to try out the product and familiarize yourself with it before committing any money to purchase;

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4. Access to source code and the ability to tweak it to suit your needs. Since you can get the source code, you can modify it, and even submit your modifications/enhancement to be peer-reviewed for efficiency and effectiveness;

5. Quick fix of bugs at a faster rate than closed source vendors. Actually, many bugs are fixed by the community before they are even reported by the users.

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HOW DO I BUILD A BUSINESS AROUND MY OPEN-SOURCE PROJECT?

To successfully build a business around your open-source project, the first thing you should know is that your project license holds the key to how your product is used or shared. Licenses have evolved a lot since the early days of open-source. Nowadays, the most popular license is MIT, which is very clear and liberal. It is also well adopted among corporations. Corporations know this license and are ok with it. In addition to cloud computing, it was able to open the doors for monetization in a more SaaS/PaaS-like model. It is more scalable, and it is the reason investors are interested in open-source more than ever before.

To successfully build a business around your open-source project, you will consider the following phases and ensure they are well addressed:

1. Project-community fit: Your open-source project has to create a community of developers who actively contribute to the open-source codebase. When developers like your project, they tend to contribute more to it and build other products around it. This increases your adoption base;

2. Product-market fit: This is the degree to which your project satisfies strong market demand. Your open-source software is implemented and used by

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someone. You must ensure it satisfies the demand of that customer;

3. Value-market fit: You must find a value proposition that resonates with your target customer; and which they are willing to pay for.

CREATING A PRODUCT

Daily, numerous enterprises are just looking for a product that works. Why not create it for them? Before you jump at that, remember that creating, marketing and selling a product is no different in the open-source space from any other endeavor.

- Decide a name for the product. Calling a product by a different name can be an important step which leads you and your customers in the right direction;
- You need to develop tools specific to your product name, including branding, documentation, etc.;
- Determine what you want to do with the code itself. Do you want to release all of them as open-source? While you may release all of your source code, that doesn't mean you have to release your full test plans, or your QA strategies, or your particular continuous integration suite, or anything else.

Although writing good code is the starting point, building a successful open-source business is more than good codes.

THE NEW, OPEN PLATFORM MODEL

Creating a successful product, whether open-source or proprietary, is often challenging.

A starting point here is the premise that open-source

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platforms have great value, and you sell that value in the form of a certified software product.

A valuable key is to clearly state to your market what you are selling and what you are giving away for free. Importantly, anticipate and remove potential confusion, and make sure your target market understands the inherent value of your open-source product.

Remember, successful open-source platforms are very valuable, and enterprises appreciate and pay for that value.

IMPROVING PRODUCT CREATION & MANAGEMENT

Supply chain management is a critical aspect of the open-source ecosystem today. Open-source companies that are able to better manage and influence the supply chain has a competitive edge over those that are unable to.

The Open-Source Software Supply Chain
The software industry has always had the supply chain defined by software sources inside the company itself, typically from multiple engineering teams. While a small percentage of software comes from outside the company. This system was already well established until open-source became a lot more common, and everything changed.

Role of the Software Supplier
In a bid to improve your product creation and management, you have two basic choices: either build your own internal means of vetting the code and applying product management processes, or rely on an intermediary to perform that function.

Most often, companies prefer going with an intermediary to

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vet the code, perform some quality assurance engineering, and apply whatever glue code is necessary to make it work satisfactorily. This is where software distribution becomes very important, and big players like Red Hat, and Canonical have mastered the game. These companies are very important in the software supply chain.

Without these companies and their likes, companies creating products would have to move through the whole chain all by themselves: vetting, hiring in-house experts, performing gluing processes, etc.

To become successful at open-source products, you must be able to influence and manage the sundry supply chains that ultimately come together in the product creation process. Hence, you must decide what point of the supply chain you want to invest in, while also paying close attention to changes in other ecosystems upon which you are reliant.

Achieving Maximum Efficiency

A deep understanding of your supply chain, and mastering the best way to manage it, are very important things to do to be successful.

Efficient management of the software component supply chains and simultaneous creation of an efficient supply chain funnel that allows for fast iteration on a product under development will lead to immense improvement of your product creation and management processes.

BECOMING A SUPPLY CHAIN INFLUENCER

The open-source company that is great at managing and influencing the open-source supply chain will be best

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positioned to create the most innovative products.

Evaluating Supply Chains

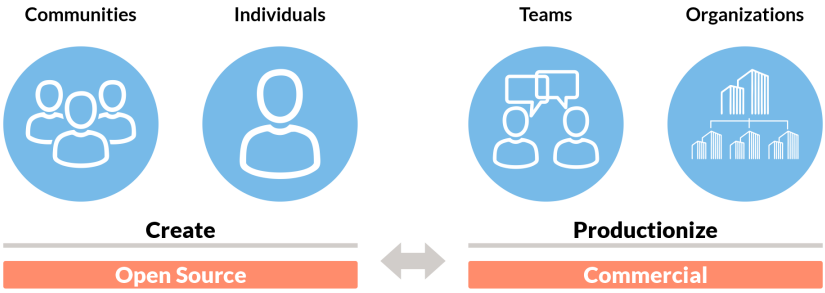
Oftentimes, in many companies, the process of selecting components that go into a product is haphazard. To determine the preferred components, your evaluation should be based on projects' longevity, stage of development, and enough other metrics to form the basis of a "build vs. buy" decision. Furthermore, the involvement of the development team in support, the number of users, and other necessary factors should be considered as well. The ability to tell when a component is becoming obsolete and then switch to something more modern is a necessary skill in the supply chain.

COMPLEMENTING OPEN-SOURCE WITH COMMERCIAL SOFTWARE

To successfully navigate this point, you may consider dividing your software into two separate versions based on the needs of your users. The open-source platform could cater to the needs of individual users, while the commercial version is targeted at the needs of businesses and teams.

This model ensures that you capture all of the values of the commercial package. This obliterates the fear of commercial open-source software vendors of cloud providers taking away their distribution business by providing hosted versions of the open-source software. Individual users derive little or no value by hosting the open-source platform by themselves. Hence, only the commercial package is worth hosting, but would not be easy since you have made it proprietary.

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CONNECTING THE USER AND CUSTOMER JOURNEYS

In connecting the user to the customer journeys, there has to be an increased awareness of open-source platforms. This can be amplified by an active and engaged community.

Awareness has to be focused on helping open-source users become aware of which aspect of their organization can benefit from open-source; and what part of the product is enterprise.

The open-source community is also made aware of the value provided by the commercial package.

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FUNDAMENTAL OPEN-SOURCE BUSINESS MODELS

OPEN-CORE

In recent times, we are beginning to see commercial companies extend their open core by building products devoid of direct conflict with their open core. This is good stewardship. In some cases, commercial companies have placed open-source project governance under the management of non-profit foundations like Apache Foundation.

Open core companies are becoming well-grounded in ensuring that the commercial offering does not conflict with the open-core, but rather complements it.

It is also important that essential functionality is not held back from the core product. Ensuring the success of the open-source is a precondition for the success of the commercial offering.

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PROFESSIONAL SERVICES (PROSERV)

Early open-source companies often had a model built around professional services being offered to companies as paid support and consultancy. Though some were able to scale with this model, it had significant challenges. Chief among these challenges was the unpredictability of services revenue. Its margins were often low in comparison to product-based companies.

HOSTING

Hosting has enabled users to use infrastructural components without having to worry about the operational overhead of managing the infrastructure. The business of hosting among open-source companies is based upon charging a price lower than that which it will cost users if they decide to host it themselves. If the price is significantly higher than the cost of the underlying infrastructure, then companies will choose to host for themselves, this is especially true for larger customers who already have sophisticated in-house DevOps teams.

MARKETPLACES

Though still a new model, being an intermediary between parties who interact with your product is a model being explored by open-source companies. An example worth considering is Mozilla generating the bulk of its annual revenue by providing lead generation to search engines.

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HOW TO PICK THE RIGHT MODEL?

In reality, companies often use a mix of all the models.

A common practice among successful open-source companies today is to possess a combination of an open-core product, hosting, and services as primary, secondary, and tertiary revenue streams respectively. To implement this model for your product, you must give it good thought on how to differentiate open-source and commercial offerings.

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A LOOK AT THE OPEN-SOURCE MARKET

MONGODB

- Founding year: 2007;
- Funding Summary: Raised a total of $311M from 6 funding rounds; IPO was in autumn 2017; valuation $1.6B;
- Started with Open-Source (license): Began with AGPL;
- Open-Source Evolvement: Built SSPL in 2018 leading to much debate in the community. SSPL is not an open-source license;
- Devtool: Database;
- Open to contributions / CLA: MongoDB owns 100% of the IP; 99.9% was developed in-house and the few external contributors signed a CLA;
- HQ: U.S.-based;
- Notes / Story synopsis: According to MongoDB, the adoption rate increased after the license change (15 million downloads more than the total downloads of the prior 10 years). In 2016 their database-as-a-service offering was launched. Until Oct. 2017 MongoDb witnessed more than 30M downloads, with 10M from

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the prior 21 months.

DATABRICKS

- Founding year: 2013;
- Funding Summary: Total funding to date: 1.9B; last round: Series G; Feb 2021- $1B;
- Started with Open-Source (license): Started with proprietary PaaS;
- Open-Source Evolvement: Though they use a lot of open-source software and are major contributors, their main service is proprietary;
- Devtool: Backend;
- Open to contributions / CLA: N/A;
- HQ: U.S.-based;
- Notes / Story synopsis: Databricks' main offering is proprietary but based on open-source. They are original creators of some popular open-source data technologies.

ELASTIC

- Founding year: Elasticsearch released in 2010; Elasticsearch Inc in 2012;
- Funding Summary: Total funding $162M; IPO in autumn 2018;
- Started with Open-Source (license): Apache 2

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open-source license for Elasticsearch;

- Open-Source Evolvement: The license was changed in 2021: Users can opt for the proprietary elastic license or SSPL;
- Devtool: Devtool;
- Open to contributions / CLA: CLA;
- HQ: U.S.-based;
- Notes / Story synopsis: In 2018, elastic (all their products) IPO witnessed its shares double on the first day of trading.

CONFLUENT

- Founding year: 2011;
- Funding Summary: Last round: Series E. Total funding- $455.9M;
- Started with Open-Source (license): The Confluent Community License is not open-source and has a few restrictions, unlike Apache Kafka available under the Apache 2.0 license;
- Open-Source Evolvement: Confluent is not open-source, but Kafka is open-source;
- Devtool: Devtool;
- Open to contributions / CLA: N/A;
- HQ: U.S.-based;
- Notes / Story synopsis: Confluent was founded by the same team who created Apache Kafka. The team has

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made lots of contributions to open-source prior to the creation of Confluent. The Confluent code itself is not open-source, but relies heavily on other open-source software for its tech stack.

REALMDB

- Founding year: 2011;
- Funding Summary: They had four investment rounds before their acquisition by MongoDB in April 2019 for $39M;
- Started with Open-Source (license): Began as proprietary; then switched over to open-sourced and adopted the open core model. Subsequently, they open-sourced the Sync solution too, opting for the hosted (SaaS) model;
- Open-Source Evolvement: Switched from proprietary to open core to open SaaS; acquired by MongoDB to push their backend offerings and complement it with an edge and sync to serve Mobile and IoT better;
- Devtool: Database;
- Open to contributions / CLA: Accepted contributions;
- HQ: Moved HQ from Europe to the U.S., but now owned by MongoDB;
- Notes / Story synopsis: Prior to their acquisition by MongDB, both founders exited. The acquisition price paid by MongoDB was less than what Realm had raised in the years before. The Sync solution is now tied to using the Mongo servers and is a huge part of their push for the IoT market.

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SQLITE

- Founding year: 2000;
- Funding Summary: Bootstrapped;
- Started with Open-Source (license): Public Domain, i.e, open-source;
- Open-Source Evolvement: Public Domain, monetization comes from being in a consortium with big corporations. Also offers paid services;
- Devtool: Database;
- Open to contributions / CLA: 99% of IP developed and owned by Richard Hipp; very limited external contributors with no rights;
- HQ: U.S.-based;
- Notes / Story synopsis: The company is still being managed by the founder and his wife, together with about three part-time workers for special assignments.

COUCHBASE LITE

- Founding year: 2011 — Couchbase, Inc. is a merger of Membase and CouchOne in 2011; both former companies were started in 2009 and had funding;
- Funding Summary: Latest Series G for $105 million; the total amount raised was: $251M;
- Started with Open-Source (license): Apache 2;

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- Open-Source Evolvement: Delayed Open-Source;
- Devtool: Database;
- Open to contributions / CLA: N/A
- HQ: U.S.-based;
- Notes / Story synopsis: Server and cloud services through the sale of Couchbase Servers are now the main business of Couchbase. Couchbase Lite is the smallest part of their business.

REDIS

- Founding year: 2009;
- Funding Summary: Total funding: $246.6M;
- Started with Open-Source (license): Switched from APGL to Apache 2.0 with Common Clause to a proprietary license called “Redis Source Available License”;
- Open-Source Evolvement: Redis itself is BSD; RedisLabs licenses its features and extensions around Redis under proprietary licenses;
- Devtool: Database;
- Open to contributions / CLA: All contributions need a CLA provided by redislabs;
- HQ: Redislabs is U.S.-based. Salvatore Sanfillipo (antirez) was based in Europe; Redislabs originated in Israel;
- Notes / Story synopsis: Redis is marketed by the

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commercial company RedisLabs; Redis was largely developed by Salvatore Sanfilippo. He left Redis in 2020.

RED HAT

- Founding year: 1993;
- Funding Summary: Total amount raised was $240.7M; acquired by IBM in 2019 for $34 billion;
- Started with Open-Source (license): Linux, which was the core of the success of Red Hat, is GPL;
- Open-Source Evolvement: Red Hat is a successful commercialization of open-source, which does not necessarily fit this matrix;
- Devtool: Backend / Data centric;
- Open to contributions / CLA: Contributions are permitted without a CLA;
- HQ: U.S.-based;
- Notes / Story synopsis: Red Hat has the majority of its success based on Linux. They even offered option shares to Linus Torvald (creator of the Linux kernel) out of gratitude.

MYSQL

- Founding year: 1995;
- Funding Summary: Total funding raised: $39.8M; acquired by Sun for $1B in 2008;

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- Started with Open-Source (license): Began with AGPL;
- Open-Source Evolvement: Dual Licensing and Paid Support;
- Devtool: Database;
- Open to contributions / CLA: Accepts contribution even though called OCA (Oracle Contributor Agreement);
- HQ: Headquartered in Sweden until its acquisition by Sun Microsystems in 2008 (who then were acquired by Oracle);
- Notes / Story synopsis: Most of the original database code base was developed by Michael Widenius.

HYPER

- Founding year: 2010;
- Funding Summary: Undisclosed;
- Started with Open-Source (license): Proprietary, not open-source;
- Open-Source Evolvement: None;
- Devtool: Database;
- Open to contributions / CLA: N/A;
- HQ: E.U.-based;
- Notes / Story synopsis: 2016: Hyper was acquired by Tableau in 2016 on undisclosed terms.

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PARSTREAM

- Founding year: 2011;
- Funding Summary: ParStream was acquired by CISCO in 2015;
- Started with Open-Source (license): Proprietary, not open-source;
- Open-Source Evolvement: N/A;
- Devtool: Database;
- Open to contributions / CLA: N/A;
- HQ: Initially headquartered in the E.U. (German), relocated to the U.S. in 2012, acquired by Cisco in 2015;
- Notes / Story synopsis: The functionality of Cisco ParStream is now part of Cisco Kinetic.

COCKROACH DB

- Founding year: 2015;
- Funding Summary: Had Series E funding round in Jan 2021 for $160M;
- Started with Open-Source (license): Apache 2.0, plus a proprietary license for enterprise features;
- Open-Source Evolvement: It began as open core, then evolved into a form of closed SaaS with delayed open-source. They switched to a proprietary license in 2019, called BSL, which prohibits users from offering

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CockroachDB as a service (DBaaS, SaaS), but each release becomes an open-source license three years after its release;

- Devtool: Database;
- Open to contributions / CLA: CockroachDB has had over 1590 commits from over 320 external contributors across all our open-source repositories;
- HQ: U.S.-based;
- Notes / Story synopsis: CockroachDB changed its license from Apache License 2.0 (free software license) to Business Source License (BSL) which is their own proprietary license. It forbids offering a commercial version of CockroachDB as a service without a paid license; while still accessible for free by the community.

BERKELEY DB

- Founding year: 1994;
- Funding Summary: Oracle acquired it in 2006;
- Started with Open-Source (license): BSD and Sleepycat Public License (a permissive OSS license);
- Open-Source Evolvement: Oracle changed to dual licensing, offering APGL and a commercial license;
- Devtool: Database;
- Open to contributions / CLA: N/A;
- HQ: U.S.-based;

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- Notes / Story synopsis: Berkeley DB is still used in many routers.

GITHUB

- Founding year: 2008;
- Funding Summary: It was bought by Microsoft In 2018 for $7.5 billion;
- Started with Open-Source (license): Proprietary;
- Open-Source Evolvement: N/A;
- Devtool: Backend / Data centric;
- Open to contributions / CLA: N/A;
- HQ: U.S.-based;
- Notes / Story synopsis: The proprietary model has worked for them since after their acquisition by Microsoft.

GITLAB

- Founding year: Its development began in 2011; it was later incorporated in 2014;
- Funding Summary: Series E $434.2M;
- Started with Open-Source (license): Open-source (MIT license);
- Open-Source Evolvement: It is an Open Core Model; Its Community Edition is MIT License; Its Enterprise

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Edition is source-available proprietary software;

- Devtool: Backend / Data-centric;
- Open to contributions / CLA: Majorly Apache 2.0.;
- HQ: U.S.-based;
- Notes / Story synopsis: The codebase of the core system was by and large developed by one person.

MARIADB

- Founding year: 2009;
- Funding Summary: Total funding raised: $123.2M;
- Started with Open-Source (license): It has dual licensing with GPL version 2 license, and a proprietary source available license for some other parts;
- Open-Source Evolvement: They now use the proprietary source available license (BSL);
- Devtool: Database;
- Open to contributions / CLA: Yes it is open to contributions, and the CLA is shared under a creative commons license that allows users freedom;
- HQ: Sweden-based;
- Notes / Story synopsis: MariaDB has about 20m users, over €100m backing; and a growing database business. All of these happened 10 years after it was forked.

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CAN YOU MAKE A BUSINESS OF OPEN-SOURCE HARDWARE?

Given the success of open-source software, there have been efforts to create similar business success in open-source hardware. Open-Source Hardware (OSHW) is hardware whose design is made publicly available so that anyone can study, modify, manufacture, distribute, and sell the design or pieces of hardware based on that design (OSHWA, 2016).

Initially, it was believed that open-source hardware is challenging to the nature of design necessary to build physical objects, however, with modern advancements in digital manufacturing technology these views have changed.

SparkFun, Adafruit, Arduino, Olimex, DIY Drones, LulzBot, Pinoccio are successful open-source hardware companies leading the way.

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WAYS TO MAKE A LIVING WITH OPEN-SOURCE HARDWARE

SELL YOUR OWN PRODUCTS

Selling your own product is the most popular business model for open-source hardware creators. Many successful companies use this model. Since open-source is about taking from each other, much successful open-source hardware influence one another's designs, from which they make their products.

SELL PRODUCTS MADE BY OTHERS

It is noteworthy that many open-source hardware companies started off as retail shops selling products made by others before adding their own products.

Even other successful open-source hardware companies sell other companies' products.

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SELL PRODUCTS MADE WITH OTHERS

The idea behind open-source hardware is that products are designed as a collaborative effort between you and others. For example, Inventables started as an online retail store (#model 2) and helped Shapeoko inventor go to market by working with him on the design, branding, manufacturing, and distribution.

SELL A SERVICE

Some open-source hardware companies have their focus on providing services to other hardware companies. Examples of such companies include SeeedStudio and OSH Park.

SELL YOUR EXPERTISE

Equipped with your expertise in a subject area, you can become a consultant, and get paid to advise other companies. To monetize your expertise, you may even organize paid events like conferences, workshops, coaching, etc., though this model is well established in the open-source software industry, it is still new in the open-source hardware industry.

SELL WORKSHOPS

Organizing paid workshops to teach people how to build products can become a key activity for your organization. Some open-source hardware projects prefer to organize workshops for free, asking people only to buy the kit. Creating strong and delightful experiences is very important

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for this model to work.

ASK FOR DONATIONS

Technology has made us capable of asking for donations not only from friends and family, but also from total strangers through crowdfunding. You could also add a donation button/link on your website, and perhaps also sell merchandise around your brand.

BECOME PART OF ANOTHER COMPANY

You could enter into a partnership with a bigger company for a manufacturing or distribution deal. Some open-source hardware projects even start as an offshoot from bigger companies.

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THE CHALLENGE OF OPEN-SOURCE BUSINESS MODELS

The use of open-source software has grown over the decades with massive success, there are still some sticky points to watch.

1. Security: The risk of a security breach is always present with the use of open-source software. An IT security professional should be consulted before committing to an open-source platform. Safeguards could be installed to minimize the risk of a cyberattack;

2. Updates: Open-source software often does not have any particular vendor releasing updates. Hence, developers must find them. Business leaders often need to implement company structures that require IT teams to manage patch through and updates to ensure they remain secure and functional;

3. Lack of Customer Support: The lack of customer support implies that users have to seek out online forums to look for solutions to their problems with the software. If you have also been able to solve any particular problem you encountered with the software, kindly share on the forum to help someone;

4. Compatibility: Compatibility issues may arise when some proprietary software are unable to work with

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open-source ones. At other times, a company's IT professionals may be unfamiliar with the concerned software. The solution is to commit the necessary resources to make it work;

5. Training: Open-source software does not usually come with a suite of training resources, as is the case with paid-and-packaged software. To mitigate this, you should have a training strategy for your team;

6. Community and Licensing: It is important that your choice of open-source software has an engaged and active community. It is also important that you understand the content of the license and ensure it aligns with your business goals and objectives;

7. Confusing Complexity: An open-source software may have complex designs, which makes its functionality unnecessarily challenging for users. An in-house developer can help work through all of the open-source work and integrate them appropriately with the needed tasks;

8. Not Realizing the Cost: A common pitfall is that companies do not often calculate the cost of running and maintaining open-source software. It sometimes takes a ton of time to manage open-source platforms. To avoid this, a comparison should be done between the net cost of open-source software and commercial alternatives to ensure appropriate decisions;

9. Not Prioritizing a Policy: Your company should have a policy document guiding its use of software, both open-source, and commercial. A well-thought-out policy document will guide against many software issues like incompatibilities;

10. Learning Curve: Users who are not technologically

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inclined could have tons of issues learning how to use open-source software. There are numerous online tutorials on how to use various software products. Check them out and try your hands on the software;

11. Mystery Sources: Independent developers can make changes to the software, which could cause problems for your business. This could pose some serious risks to your online privacy and safety;

12. Seeing the Big Picture: Despite all the advantages of open-source software, the complex nature of software could make it hard for you to step back and take a look at the big picture. Continuous practice with the software will increase your proficiency and ability to view your results with the big picture in mind.

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CRITICISM

1. Revenue Limits: Since most open-source companies cannot monetize their products effectively, then they cannot effectively invest substantially in innovation. This implies that potential corporate clients can not benefit from the fruits of that innovation, and that's not so good for customers;

2. Uneven Playing Field: Open-source companies compete both with their freely available software on which their products are built and with similar products sold by proprietary software companies. The field is uneven, as the low revenues from the open-source companies are no match for the humongous earnings of their commercial counterparts;

3. Hybrid Future: The most feasible way for open-source software companies to make money is to abandon the standard open-source business model of selling support and maintenance subscriptions, and, instead, using open-source software as a platform on which to build software as a service (SaaS) offerings.

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CONCLUSION

Open-source software is no longer for hobbyists. It is now a sophisticated, multi-billion dollar business industry. It has evolved over the decades into a business strategy with broad applicability. Businesses are being built around this idea.

Not all open-source software has the same profit potential. Nonetheless, the most successful open-source software businesses have successfully employed a mix of all the discussed business models here.

Given the current trajectory, the future of open-source software is bright. Increasingly, open-source software will play a bigger role in our daily lives. This presents even bigger opportunities for businesses.

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